

# Math 39 Course at a Glance

## Precalculus with Data Analysis

Topical units are not numbered because instructional order may vary. Objectives are grouped by curriculum priority; supporting skills are intentionally omitted from this summary.

### Linear Modeling and Residuals

#### REQUIRED

|                    |                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------|
| <b>M39-LIN-001</b> | I can calculate and interpret average rate of change with appropriate units.                                     |
| <b>M39-LIN-002</b> | I can construct a linear function from two data points, a table, a graph, or a context.                          |
| <b>M39-LIN-003</b> | I can fit a linear regression model to data using technology and interpret its parameters.                       |
| <b>M39-LIN-004</b> | I can interpret a correlation coefficient as evidence about the direction and strength of a linear relationship. |
| <b>M39-LIN-005</b> | I can calculate residuals, construct a residual plot, and use its pattern to evaluate a model.                   |
| <b>M39-LIN-006</b> | I can use a linear model to make and interpret predictions, including solving for either variable.               |

### Exponential Modeling

#### REQUIRED

|                    |                                                                                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------|
| <b>M39-EXP-001</b> | I can distinguish linear and exponential change using rates, multipliers, tables, graphs, or contexts.       |
| <b>M39-EXP-002</b> | I can construct an exponential function from two data points, a table, or a context.                         |
| <b>M39-EXP-003</b> | I can interpret the initial value, growth or decay factor, and growth or decay rate of an exponential model. |
| <b>M39-EXP-004</b> | I can determine and interpret doubling time or half-life from an exponential model.                          |
| <b>M39-EXP-005</b> | I can solve exponential and logarithmic equations in modeling contexts.                                      |
| <b>M39-EXP-006</b> | I can fit an exponential regression model using technology and evaluate its fit.                             |
| <b>M39-EXP-007</b> | I can compare linear and exponential models and justify which is more appropriate for a data set or context. |

### Power Functions and Function Comparison

#### REQUIRED

|                    |                                                                                                                    |
|--------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>M39-POW-001</b> | I can identify a power function and interpret the parameters in $y = kx^p$ .                                       |
| <b>M39-POW-002</b> | I can construct a power function from two data points or a proportional relationship.                              |
| <b>M39-POW-003</b> | I can relate the sign and size of $k$ and $p$ to the shape, domain behavior, and end behavior of a power function. |
| <b>M39-POW-004</b> | I can recognize and model direct, inverse, and other power-variation relationships.                                |
| <b>M39-POW-005</b> | I can solve equations involving integer or rational powers.                                                        |
| <b>M39-POW-006</b> | I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.                 |
| <b>M39-POW-007</b> | I can fit a power regression model and interpret its parameters and correlation evidence.                          |

### Function Combinations and Polynomials

#### REQUIRED

|                    |                                                                                                                    |
|--------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>M39-POL-001</b> | I can add, subtract, multiply, and divide functions represented by tables, graphs, or formulas.                    |
| <b>M39-POL-002</b> | I can determine where a combination of functions is positive, zero, or undefined.                                  |
| <b>M39-POL-003</b> | I can determine polynomial end behavior from degree and leading coefficient.                                       |
| <b>M39-POL-004</b> | I can identify zeros and multiplicities and relate them to polynomial graph behavior.                              |
| <b>M39-POL-005</b> | I can estimate a polynomial's possible degree from its zeros, turning points, inflection points, and end behavior. |
| <b>M39-POL-006</b> | I can construct a polynomial function from specified zeros, multiplicities, and a point.                           |
| <b>M39-POL-007</b> | I can match polynomial formulas and graphs using intercepts, multiplicity, degree, and end behavior.               |
| <b>M39-POL-008</b> | I can build and optimize a polynomial model in a geometric or economic context.                                    |

# Math 39 Course at a Glance

## Precalculus with Data Analysis

### Describing Data

#### REQUIRED

|                     |                                                                                                    |
|---------------------|----------------------------------------------------------------------------------------------------|
| <b>M39-ST A-001</b> | I can distinguish categorical and quantitative data.                                               |
| <b>M39-ST A-002</b> | I can construct and interpret frequency and relative-frequency tables.                             |
| <b>M39-ST A-003</b> | I can construct and interpret bar charts, pie charts, and histograms using appropriate technology. |
| <b>M39-ST A-004</b> | I can identify misleading features in a data display.                                              |
| <b>M39-ST A-005</b> | I can describe a distribution's modality, symmetry, skewness, and outliers.                        |
| <b>M39-ST A-006</b> | I can calculate and interpret mean, median, and mode and select an appropriate measure of center.  |
| <b>M39-ST A-007</b> | I can calculate and interpret range, standard deviation, quartiles, and interquartile range.       |
| <b>M39-ST A-008</b> | I can construct and interpret a five-number summary and boxplot.                                   |
| <b>M39-ST A-009</b> | I can use percentiles and z-scores to locate and compare values within distributions.              |
| <b>M39-ST A-010</b> | I can compare distributions using their shape, center, spread, and unusual features.               |

### Probability

#### REQUIRED

|                    |                                                                                                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>M39-PRO-001</b> | I can organize categorical data in a contingency table or Venn diagram.                                           |
| <b>M39-PRO-002</b> | I can calculate and interpret empirical, marginal, joint, and conditional probabilities from a contingency table. |
| <b>M39-PRO-003</b> | I can identify an experiment, outcomes, events, and a sample space.                                               |
| <b>M39-PRO-004</b> | I can calculate theoretical probabilities when outcomes are equally likely.                                       |
| <b>M39-PRO-005</b> | I can use complements to calculate probabilities.                                                                 |
| <b>M39-PRO-006</b> | I can calculate probabilities of compound 'and' events and distinguish independent from dependent events.         |
| <b>M39-PRO-007</b> | I can calculate probabilities of compound 'or' events and distinguish overlapping from disjoint events.           |
| <b>M39-PRO-008</b> | I can compare experimental and theoretical probability and explain the law of large numbers.                      |

### Expected Value

#### EXTENSION

|                   |                                                                                |
|-------------------|--------------------------------------------------------------------------------|
| <b>M39-EV-001</b> | I can calculate the expected value of a discrete probability situation.        |
| <b>M39-EV-002</b> | I can interpret expected value to evaluate a game, risk, or long-run decision. |

### Linearization of Nonlinear Data

#### EXTENSION

|                    |                                                                                                     |
|--------------------|-----------------------------------------------------------------------------------------------------|
| <b>M39-POW-008</b> | I can linearize power or exponential data and translate between the linearized and original models. |
|--------------------|-----------------------------------------------------------------------------------------------------|